



Semester 2 Examinations 2015/2016

Exam Code(s) 3BME1, 3BMS2, 3BPP1, 3BS9, 3FM2, 4BMS2, 4BS2
Exam(s) 3rd Arts, 3rd & 4th Science

Module Code(s) MP307, MP341
Module(s) Modelling II, Mathematical Modelling

Paper No. 1
Repeat Paper No

External Examiner(s) Professor P. Martin
Internal Examiner(s) Dr P. Piironen

Instructions: Marks are awarded for question 1 in Section A **and** the TWO best questions in Section B.
All questions carry the same weight.

Duration 2 hours
No. of Pages 6
Discipline(s) Applied Mathematics
Course Co-ordinator(s) Dr P. Piironen

Requirements:

Release in Exam Venue Yes ☒ No ☐

Handout None
Statistical/ Log Tables Yes
Graph Paper None
Log Graph Paper None
Other Materials Calculator

Section A - Answer all questions (a)-(d). (Each question is worth **5 Marks**.)

1. (a) Consider a finite queue of maximum length n and let

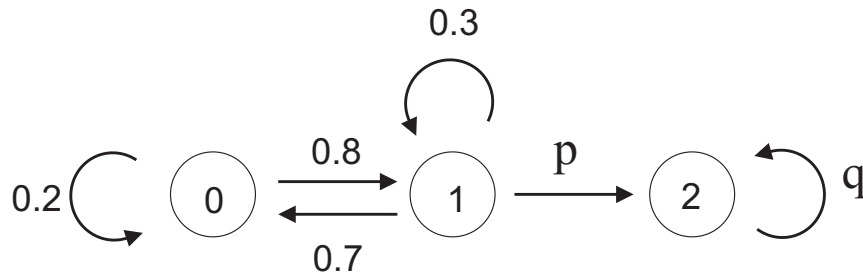
$$p_k = \frac{1}{n+1}, \quad k = 0, 1, 2, \dots, n \quad (\text{Uniform distribution})$$

be the probability that the queue is of length k .

- (i) What is the expected queue length (the mean)?
(ii) What is the variance and the standard deviation of the queue length?
You may use that

$$\sum_{k=0}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

- (b) Consider the following Markov Process with transition probabilities shown.



- (i) Find the transition matrix P and the probabilities p and q .
(ii) Is the Markov process ergodic? Explain your answer.
(iii) Does the Markov process have unique equilibrium probabilities? Explain your answer. If the Markov process has unique equilibrium probabilities calculate them.
- (c) Consider a model of two competing species P_1 and P_2 given by

$$\begin{aligned} \frac{dP_1}{dt} &= 40P_1 - 4P_1^2 - 4P_1P_2, \\ \frac{dP_2}{dt} &= 60P_2 - 3P_1P_2 - 3P_2^2. \end{aligned}$$

- (i) Give an interpretation of the different terms in the model.
(ii) Find all equilibrium points for P_1 and P_2 .

(d) Consider the linear control system

$$\begin{aligned}\frac{dx_1}{dt} &= x_2, \\ \frac{dx_2}{dt} &= bx_1 + x_2 + u, \\ y &= x_1 + x_2,\end{aligned}$$

where $x_1 = x_1(t) \in \mathbb{R}$ and $x_2 = x_2(t) \in \mathbb{R}$ represent the state, $u = u(x_1, x_2) \in \mathbb{R}$ is the input, $y \in \mathbb{R}$ is the output and $b \in \mathbb{R}$ is the control parameter.

- (i) Is this system observable? Explain your answer.
- (ii) Is this system reachable? Explain your answer.

Section B - Answer any two questions.

2. Assume that the size of a queue is given by a nearest-neighbour Markov model and that it has a Poisson arrival pattern with parameter α_k and a Poisson servicing pattern with parameter β_k , where k is the queue size.

- (a) Let the maximum size of the queue be r .
(i) Explain why $\alpha_r = \beta_0 = 0$.
(ii) Assuming that the Markov Process is ergodic, show that the equilibrium probabilities are given by

$$\pi_k = \rho_k \rho_{k-1} \dots \rho_1 \pi_0$$

for $0 < k \leq r$ where $\rho_k = \alpha_{k-1}/\beta_k$.

8 Marks

- (b) Let the queue in (a) have

$$\alpha_k = \alpha, \quad \beta_k = \beta,$$

where $\alpha > 0$ and $\beta > 0$.

Show that the equilibrium probabilities π_k , $k = 0, 1, 2, \dots, r$, are given by

$$\pi_k = \begin{cases} \frac{\rho^k(1-\rho)}{1+\rho^{r+1}}, & \rho \neq 1, \\ \frac{1}{r+1}, & \rho = 1, \end{cases}$$

where $\rho = \alpha/\beta$.

6 Marks

- (c) Let the queue in (b) correspond to a doctor's waiting room that can hold 4 people. The doctor can treat 5 people per hour and one patient arrives every 10 minutes on average. What is the probability that the waiting room is full?

6 Marks

3. Consider an insect species in which no individuals live beyond 3 days. Divide the insect population into four age groups labelled by $n = 0, 1, 2$ and 3. Assume that only the second ($n = 1$) and third ($n = 2$) groups can reproduce. For each group n , let b_n denote the birth rate and d_n the death rate.

- (a) Find a matrix $\mathbf{A}$ such that $\mathbf{P}(t+1) = \mathbf{A}\mathbf{P}(t)$ with

$$\mathbf{P}(t) = \begin{pmatrix} P_0(t) \\ P_1(t) \\ P_2(t) \\ P_3(t) \end{pmatrix},$$

where $P_n(t)$ denotes the population size of age group n at day t .

4 Marks

- (b) Show that the eigenvalues of $\mathbf{A}$ satisfy the equation

$$\lambda^4 - b_1(1 - d_0)\lambda^2 - b_2(1 - d_0)(1 - d_1)\lambda = 0.$$

4 Marks

- (c) If $d_0 = 0.1$, $d_1 = 0.2$, $d_2 = 0.4$, and $b_1 = 0.7$, $b_2 = 0.6$, show that there is one eigenvalue greater than 1. What is the significance of this?

7 Marks

- (d) Verify your conclusion in (c) by considering this case over three days, where the initial population is $P_0 = P_1 = 100$ and $P_2 = P_3 = 50$.

5 Marks

4. A two-compartment control system for drug delivery can be modelled as

$$\begin{pmatrix} \dot{c}_1 \\ \dot{c}_2 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} 0 \\ 3 \end{pmatrix} u, \quad y = c_1,$$

where c_1 and c_2 are the concentrations of the drug in each compartment, u is the flow rate of the drug into compartment 2 and y is the concentration of the drug in compartment 1.

(a) Investigate if the system is observable and/or reachable.

8 Marks

(b) Design a control function

$$u = -Kc + k_r r, \quad \text{for } K = \begin{pmatrix} k_1 & k_2 \end{pmatrix}, \quad c = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

where k_1, k_2, k_r are the control parameters, such that the desired output $y = r = 1$ and the closed loop eigenvalues for the system are $\lambda_1 = -1$ and $\lambda_2 = -2$.

12 Marks